

HwaCom: Realizing digital transformation with highest- possible standards of information security

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

ABOUT HWA.COM



About HwaCom

As a part of its digital transformation plan, HwaCom leverages its Workspace and Cloud security solutions to enhance work collaboration efficiency and reduce maintenance work.

Established in 1994, HwaCom is a leading broadband system integrator in Taiwan, offering services across information communication technology, digital media, intelligence systems and information security. Currently serving more than 830 enterprise customers across multiple industries, HwaCom embarked on digital transformation in late 2020 to provide cutting-edge technology solutions.

Industries: Technology

while meeting high information security standards.

Location: Taiwan

Storage (<https://cloud.google.com/storage/>),



Google Cloud results

- Enables flexible file storage and sharing in compliance with information security requirements through Drive
- Simplifies internal communication and managements with Forms and Sheets
- Supports live corporate events online with more than 700 employees across eight

Reduced
main work
70%

About Microfusion

Microfusion is a Taiwanese cloud service provider founded in 2007 focused on offering one-stop solutions and consulting to help customers embark on cloud migration. Having worked with more than 2,000 customers from public and private sectors, Microfusion was selected by CIO Magazine as one of the top 10 cloud service providers in the Asia-Pacific region in 2020.

Google Cloud (<https://cloud.google.com/>) service provider founded in 2007 focused on offering one-stop solutions and consulting

Google Cloud Armor
(<https://cloud.google.com/armor/>)

Cloud DNS (<https://cloud.google.com/dns/>) public and private sectors, Microfusion was selected by CIO

Cloud Storage (<https://cloud.google.com/storage/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Google Workspace
(<https://workspace.google.com/>)

office locations

via Meet

- Enhances connection stability between internal and external networks through Cloud DNS

Founded in 1994, HwaCom is a broadband system integrator that quickly became an industry leader in Taiwan, with nationwide projects including the 3G and 4G network infrastructure of the country's largest telecommunications company, as well as the traffic management and control systems of the major highways in Taiwan.



Currently offering services ranging from information communication technology to intelligence systems and information security, HwaCom has more than 830 enterprise customers across multiple industries. The company noticed that along with digitalization, customers now demand more software and managed services than hardware. To meet its customers' needs and develop more innovative products with the latest technologies like cloud, the company decided to embark on a digital transformation journey in late 2020.

“We’re truly
impressed by
how
comprehensive
the services of
Google
Workspace are.
With Google
Workspace,
we’re able to not
only store and
manage our
internal files
following
international

information
security
standards, but
also replace our
existing
scattered work
coordination
tools with a one-
stop solution.”

—Jademon Kuan, VP,
HwaCom

As a part of its digital transformation plan, HwaCom needed to optimize its internal coordination and management systems. The company had been relying on several software providers for its internal file storage, email, and calendar systems, which were not fully synchronized and required regular maintenance. During an external audit in 2020, the company was also advised to strengthen the control of its internal file access control and file sharing, with declaration and stronger encryption. As a system integrator of ISO 27001 and ISO 27701 information security certifications, HwaCom needed a new solution for internal coordination in management to help continue providing customers with ease of mind that their services meet high security standards.

To build an internal file storage system that can be better controlled, HwaCom first moved its files to an on-premises private cloud. But the team soon found it challenging to manage authorization and access of more than 700 employees in various departments on its own. The company then started to evaluate public cloud storage solutions, and eventually landed on [Google Workspace](https://workspace.google.com/)

(<https://workspace.google.com/>) in early 2021 for its unlimited cloud storage, easy authentication and authorization management, as well as the fact that it aggregates email, calendar, video meeting, and other workplace collaboration tools in one place.

“We’re truly impressed by how comprehensive the services of Google Workspace are,” says Jademond Kuan, VP at HwaCom. “With Google Workspace, we’re able to not only store and manage our internal files following international standards for information security, but also replace our existing scattered work coordination tools with a one-stop solution.”

Optimizing performance with a synchronized and secure work coordination platform

Prior to the migration, HwaCom had to transfer its existing employees’ accounts to Google Workspace, so that it would require less effort in credentials management. To streamline its employees’ login process while ensuring maximum security, the company also needed to combine the user verification method of its VPN that employees use to connect to the company’s systems from external

networks, with the 2-step verification mechanism of Google Workspace. For implementation, Google Cloud partner Microfusion (<https://www.microfusion.cloud/en/>), proposed the most ideal verification process design and provided timely technical support. With their assistance, HwaCom swiftly adopted Google Workspace in one month.

HwaCom's employees now use Google Drive (<https://workspace.google.com/products/drive/>) to save internal documents as backup and to share files internally, which allows the MIS team to easily monitor the use and sharing of files for information security audits. The previous email system only allowed HwaCom's employees to check their emails on a desktop and frequently reached the storage limit. With Gmail (<https://workspace.google.com/products/gmail/>), they can now login to their mailbox on multiple devices while enjoying unlimited storage, securely. The synchronization feature of Google Calendar (<https://workspace.google.com/products/calendar/>) and the collaborative editing feature of Docs (<https://workspace.google.com/products/docs/>), Sheets (<https://workspace.google.com/products/sheets/>) and Slides (<https://workspace.google.com/products/slides/>) have also helped enhance the company's teamwork efficiency by greatly simplifying the way of scheduling meetings and facilitating cross-department collaboration.

Before the adoption of Google Workspace, HwaCom's MIS team spent a considerable amount of time on maintaining its work collaboration

systems and resolving related issues such as system crashes and suspicious emails. Thanks to the fully managed services of Google Workspace and its email filtering and virus scanning capabilities, the maintenance workload of the MIS team has been reduced by over 70%, enabling the team to invest more resources in strengthening information security and developing ways to expand the company's application of Google Workspace tools.

"Google Workspace has significantly changed the way our employees coordinate their work and allowed them to collaborate remotely at ease. When the Taiwan government in May 2021 requested corporate employees to work remotely for two months following a new COVID-19 wave, our employees were able to quickly adapt to the new work norm," explains Gary Chen, CEO of HwaCom.

"The ability to connect Sheets with third-party data analytics tools and the powerful data aggregation and reporting

capabilities of
Forms, has
enabled us to
enhance our
productivity and
management
efficiency, which
is especially
helpful in a
remote work
setting.”

—Jerry Yang, Senior
Director, HwaCom

Enhancing productivity and management efficiency with creative use of Google Workspace

Besides daily work coordination, HwaCom also takes full advantage of Google Workspace to enhance productivity and management efficiency by combining the Google Workspace tools with its other systems. For example, the company’s sales department used to work on Excel sheets to key in the information of potential customers, but the lack of data examination often led to input errors and prolonged review time. By leveraging Forms (<https://workspace.google.com/products/forms/>) to create a report system that requests sales personnel to complete all the predefined fields, the

input can then be sent immediately by email to relevant contact points in other departments. This switch has shortened the new customer report process by one week.

To facilitate work management, HwaCom's Products Management and information Security service support department inputs their work status on Sheets and uses a data visualization tool to build dashboards. Managers can then closely follow the latest work distribution and rearrange workloads if necessary. During the COVID-19 lockdown, HwaCom used Sites

(<https://workspace.google.com/products/sites/>) to create a portal that contained all departments' weekly work reports generated by Sheets, which enabled its executives to easily track work progress.

The way the company runs meetings also experienced a huge shift. Previously, HwaCom could only provide one online video meeting at a time to each of its departments. The capacity of this system became insufficient when some employees started working from home. With Meet

(<https://workspace.google.com/products/meet/>), HwaCom's employees are able to launch online meetings any time, greatly enhancing work efficiency. In January 2021, the company even successfully held its year-end corporate event that gathered more than 700 employees across eight office locations on Meet, with the support of YouTube's live broadcast feature.

"One thing we like a lot about Google Workspace is that we can flexibly combine its tools with other

software to meet our needs,” says Jerry Yang, Senior Vice Manager at HwaCom. “The ability to connect Sheets with third-party data analytics tools and the powerful data aggregation and reporting capabilities of Forms, has enabled us to enhance our productivity and management efficiency, which is especially helpful in a remote work setting.”



Continuing cloud migration for long-term cost-effectiveness

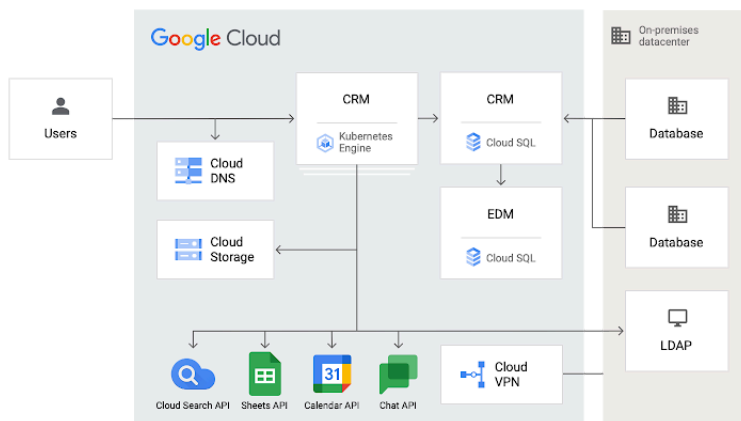
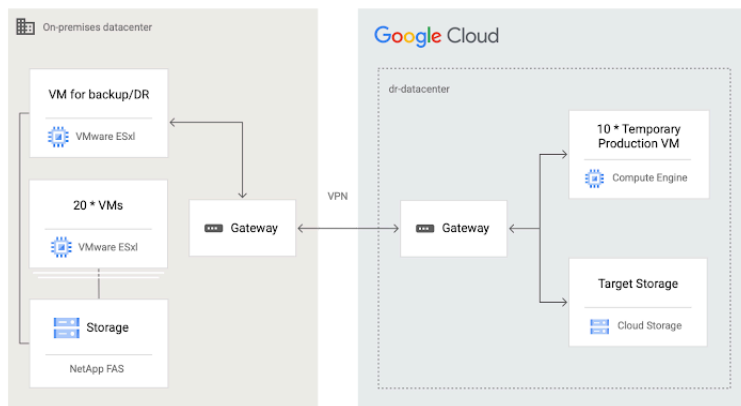
The benefits brought by Google Workspace has made HwaCom determined to accelerate its digital transformation journey. The company decided to gradually move its main management systems and some of its services to public cloud, with Google Cloud (<https://cloud.google.com/>) as the primary option because of its high compatibility with Google Workspace and the team’s positive experience of using Google Workspace.

HwaCom had a decade-old customer relationship management (CRM) system running on its server, which required its employees to connect via a

complicated process from external networks. So the company redesigned its CRM system and deployed it on [Google Kubernetes Engine](#) (/kubernetes-engine) (GKE), connected with Google Workspace, to synchronize work progress. The web-based structure of the current system has enabled HwaCom's employees to access the system through browsers, simplifying the connection process and enhancing connection stability.

HwaCom also leverages [Cloud Storage](#) (/storage) for the offsite backup required by ISO 27001 and ISO 27701 information security certifications. Yang points out that as the fee of Cloud Storage is calculated based on the use and data flow, the operational cost is significantly lower than building and maintaining an on-premises data center, because HwaCom would have needed to monitor the hardware performance on its own and constantly upgrade the infrastructure.

Furthermore, HwaCom employs GKE to develop SmartQ, a green energy distribution platform, and runs a testbed on [Anthos](#) (/anthos) to prepare for the management of Kubernetes clusters in hybrid cloud architectures that the company will adopt in the future. "Our development team will conduct development work on Google Cloud as much as possible from now on. It's more cost-effective in the long run because we don't need to renew hardware every three to five years, and our engineers can focus on product development without worrying about infrastructure maintenance," says HwaCom VP, Mingyeh Hsieh.



Strengthening network stability and information security with Cloud DNS and Cloud Armor

Dedicated to providing the most reliable and secure services to its customers, HwaCom is constantly looking for ways to strengthen its network stability and information security. The company was using its own DNS server to provide name resolution services for internal and external websites. In mid-2021, its data center experienced a power outage that lasted an entire day, which caused DNS service failure and affected external website service.

HwaCom decided to adopt Cloud DNS (/dns) and completed the migration in an hour. Since then, the DNS name resolution service has never encountered any service interruption issues.

To comply with the requirements of ISO 27001, ISO 27701, and the Taiwan Cyber Security Management Act, HwaCom also needed to set up a web-application firewall for all its web-based services to enhance information security. The company therefore set up a policy that all its website services running on GKE can only be accessed through Cloud Armor (/armor), which enables HwaCom's information security engineers to easily block access to IP addresses outside of Taiwan and prevent cyber attacks.

“After optimizing
our internal
coordination
systems, the
next step of our
digital
transformation
journey is to
share the
benefits of cloud
migration with
our customers
and partners
while ensuring
the best-

possible
information
security. The
results that
we've achieved
so far make us
convinced that
Google Cloud
can help us
reach these
higher goals.”

—Gary Chen, CEO,
HwaCom

Pursuing digital transformation with highest-possible security standards

Following the cloud migration of its CRM system, HwaCom is already in the process of moving its partner relationship management (PRM) system and enterprise resource planning system to GKE. Since many of HwaCom's outsourcing partners are smaller companies that might not have the resources to build a PRM system, the company plans to share the benefits of its cloud migration with its partners by opening access to its new PRM system on Google Cloud.

To further enhance information security, HwaCom plans to use [BeyondCorp](#) (/beyondcorp) for the access control of its core system to replace VPN, which previously took too long to establish a connection and doesn't support detailed access control. It will also leverage the standardized security log system and the threat report aggregation feature of [Chronicle](#) (/chronicle/docs) to better manage security threats.

"After optimizing our internal coordination systems, the next step of our digital transformation journey is to share the benefits of cloud migration with our customers and partners while ensuring the best-possible information security," concludes Gary Chen. "The results that we've achieved so far make us convinced that Google Cloud can help us reach these higher goals."